HYDRATION AUTOMATION (HA)

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Introduction

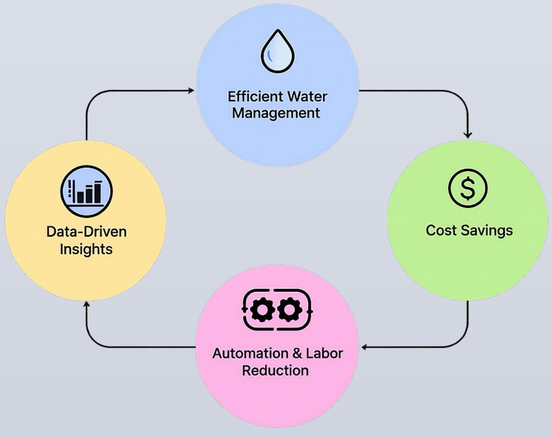
As climate change and inefficient irrigation continue to strain water resources, small community farms—often operating with limited budgets—face severe challenges. Hydration Automation (HA) is a low-cost, solar-powered, IoT-based smart irrigation system designed to address the agricultural and financial needs of small farms and nonprofits.

While similar solutions exist, they are typically cost-prohibitive for organizations with minimal resources. HA combines sensing units (SUs), LoRa communication, and a web-based dashboard to enable real-time, data-driven water management. Deployed at Veggielution, a community farm at San Jose, HA helps reduce water usage, manage limited budgets, and promote sustainable urban agriculture.

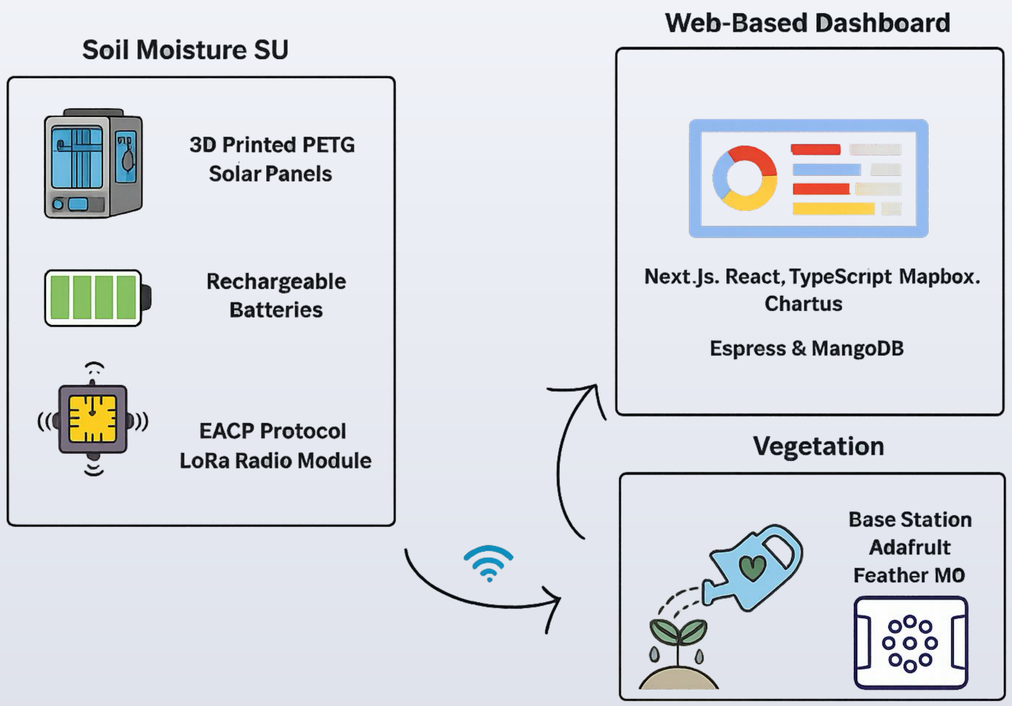
Operating Principle

HA utilizes soil moisture sensors to measure water levels. When moisture falls below a certain threshold, the automated system activates water valves, ensuring optimal hydration without human intervention.

Expected Outcomes



Project Description & Architecture



Soil Moisture Probes: Four probes per SU for cross-validation of readings.

Rechargeable Power System: Each SU uses batteries charged via integrated solar panels, supporting continuous off-grid operation.

Durable Enclosure: 3D-printed PETG casing with UV resistance, venting features, and custom-fit gaskets to withstand all environmental extremes.

LoRa Radio Modules: Enable long-range, low-bandwidth communication across multiple acres of farmland while minimizing power draw.

Energy-Aware Communication Protocol (EACP): Custom protocol dynamically adjusts transmission frequency and duty cycling based on available battery charge, ensuring longevity under energy constraints.

Wi-Fi Forwarding: Data transmitted securely from BS to back-end servers using WPA2 encryption, ensuring integrity against packet loss or tampering.

Next steps

- Optimize Sensor Hardware — Compact integration
- Enhance Data Analytics — Predictive processing
- Expand Field Deployment

Conclusion

Hydration Automation (HA) optimizes irrigation with IoT soil sensors, smart valves, and a web dashboard, reducing water waste and costs for urban farms like Veggielution. Future enhancements, including solar power and expansion, aim to revolutionize sustainable irrigation, conserving resources and promoting scalable agricultural solutions through automated water management.

Community Partner

We are partnered with Veggielution, a 6-acre urban community farm in San Jose, that promotes urban farming, youth education, and distribution of fresh produce to underserved communities. They generously provided us access to their fields for on-site testing. Their staff offer continuous feedback on how our moisture-sensor system can be adapted to real farming conditions, and help us refine our sensors, interpret soil moisture behavior, and understand how our technology can better support efficient irrigation practices.

Data-driven farming made simple — empowering urban farming initiatves

Our Team

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